

PARTS LIST

Bill of Materials

Hardware parts with indicative prices, camera selection, software, models and licenses, SBOM.

Reference installation: Lenovo ThinkCentre M73 Tiny · Ubuntu Server 26.04 LTS · ONVIF PTZ camera · Novoferm Novomatic 200 garage door operator. 100 % on-premises processing.

DOCUMENT CONTROL

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Date of issue	2026-09-30
Author	Thierry Gayet <thierry.gayet@labworks.fr>
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RECORD OF REVISIONS

Rev	Date	Description
A	2026-09-29	Initial release (French edition).
B	2026-09-30	US English edition. Content aligned with the current repository (web front end with Anubis, MQTT, remote syslog, monitoring, NTP, TLS certificate lifecycle, camera power and end-of-life data). Contents with page numbers, glossary, bibliography and index added.

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1 SCOPE AND SAFETY SUMMARY

Reference configuration: Lenovo ThinkCentre M73 **Tiny** (1-liter form factor, no PCIe slot and no DB9 port as standard), 100 % CPU inference, USB relay board, ONVIF PTZ camera on an isolated network. Prices are **indicative** (September 2026, new or refurbished for the PC) and must be checked with the supplier. "existing" = already supplied or already present. The [blue](#) links are clickable: prefilled searches at retailers (no vendor imposed) and official product pages.

Status	Meaning
REQUIRED	Needed for the base function (vision, face recognition, garage pulse).
RECOMMENDED	Strongly advised: dedicated camera network, voice, door state.
OPTIONAL	Comfort, availability or protection.

DANGER

Jarvis never switches 230 V. The relay only closes the low-voltage dry contact of terminal F of the Novomatic 200. Cut the power of the door operator before any wiring (see JRV-DWG-002).

CAUTION

PoE budget: the PTZ cameras draw up to 51 W (Axis, Hikvision) or 36 W (Dahua). A PoE+ 802.3at switch or injector (30 W) is not enough: use an 802.3bt / Hi-PoE 60 W injector or the camera's own 24 V AC / 36 V DC supply, per its datasheet.

2 PARTS LIST

2.1 Compute

Ref.	Qty	Item / specification	Role	Status	Indicative price
PC-1	1	Mini-PC Lenovo ThinkCentre M73 Tiny Machine types 10AX/10AY/10DK-10DN. Quad-core AVX2 CPU: i5-4590T, i5-4460T, i7-4785T or i7-4765T. The i5-4570T has only 2 cores: usable with detector.imgsz 416. Celeron / Pentium excluded (no AVX2). Back Market · Amazon · Lenovo PSREF sheet	Vision, voice, decision, web UI	REQUIRED	80 to 140 €
PC-2	1	Memory 2 × 4 GB DDR3 SO-DIMM 204-pin PC3-12800 8 GB recommended, 16 GB max. Dual channel = 2 identical modules. Amazon · LDLC	Models loaded in RAM (≈ 2.5 GB)	REQUIRED	0 to 20 €
PC-3	1	2.5" SATA SSD, 240 to 500 GB Replaces an original 5400 rpm hard disk if present. Amazon · LDLC	System, models (≈ 1 GB), photos, log	REQUIRED	0 to 45 €
PC-4	1	Lenovo 65 W (20 V) power adapter Normally supplied. Round or rectangular plug depending on the model. Amazon	Mini-PC power	REQUIRED	0 to 25 €

2.2 Video

Ref.	Qty	Item / specification	Role	Status	Indicative price
CAM-1	1	Outdoor ONVIF Profile S PTZ dome camera See chapter 3. Requirements: PTZ ContinuousMove / Stop / GotoPreset, secondary H.264 stream, IR or starlight, IP66. Axis Q6078-E · Axis Q6075-E (EOL) · Hikvision · Dahua	Video acquisition and tracking	REQUIRED	900 to 4,000 €
CAM-2	1	PoE injector 802.3bt / Hi-PoE 60 W, or the camera's DC/AC supply The PTZ cameras draw up to 51 W: PoE+ 802.3at (30 W) is not enough. Axis ships a High PoE 60 W midspan; Hikvision: 802.3bt or 24 V AC; Dahua: Hi-PoE or 36 V DC. Check the camera datasheet. Amazon · AliExpress	Power + data on one cable	REQUIRED	0 to 90 €
CAM-3	1	Wall / corner mount for PTZ dome Camera manufacturer's reference. Amazon	Mounting	REQUIRED	30 to 120 €

2.3 Network

Ref.	Qty	Item / specification	Role	Status	Indicative price
NET-1	1	USB 3.0 to Gigabit Ethernet adapter Realtek RTL8153 (r8152 driver) or ASIX AX88179. Amazon · LDLC	Second interface dedicated to the camera network	RECOMMENDED	12 to 25 €
NET-2	1	Outdoor Cat6 F/UTP cable, UV-resistant PE jacket Length per installation (100 m max.). Amazon	PoE link to the camera	REQUIRED	20 to 60 €
NET-3	4	Shielded Cat6 RJ45 connectors + boots To crimp, or use a pre-terminated cable. Amazon	Terminations	REQUIRED	5 to 12 €
NET-4	1	Cat6 patch cable 0.5 m USB-GbE adapter to PoE injector. Amazon	Short link	REQUIRED	3 to 6 €
NET-5	1	PoE Ethernet surge protector (RJ45, earthed) 802.3bt compatible. Amazon	Lightning protection of the outdoor cable	OPTIONAL	20 to 45 €

2.4 Audio

Ref.	Qty	Item / specification	Role	Status	Indicative price
AUD-1	1	USB microphone array Seeed ReSpeaker USB Mic Array v2.0 (XMOS XVF-3000) 4 microphones, echo cancellation, 3.5 mm jack output, direction of arrival (future use). Alternative: directional USB microphone. Seeed Studio · Seeed wiki · Amazon	Wake word, voice commands, speaker	RECOMMENDED	60 to 90 €
AUD-2	1	Class-D mini amplifier 2 × 3 W, 5 V (PAM8403 type) USB 5 V powered, input on the jack output of AUD-1. Amazon · AliExpress · datasheet	TTS amplification	RECOMMENDED	4 to 10 €
AUD-3	1	Weatherproof outdoor loudspeaker 4 to 8 Ω, 3 to 5 W IP65 or inside a protected enclosure. Amazon	Greetings and spoken answers	RECOMMENDED	12 to 30 €
AUD-4	1	Active USB 2.0 extension 5 to 10 m If the microphone is more than 3 m from the mini-PC (beyond 5 m, a passive cable is no longer reliable). Amazon	Microphone near the entrance	OPTIONAL	12 to 25 €
AUD-5	1	Weatherproof enclosure with acoustic grille For AUD-1 to AUD-3 outdoors. Amazon	Rain protection	OPTIONAL	10 to 25 €

2.5 Control and inputs

Ref.	Qty	Item / specification	Role	Status	Indicative price
IO-1	1	4-channel CH340 USB relay board "LCUS-4" USB 1a86:7523 → /dev/jarvis-relay. 10 A contacts. Protocol A0 channel state checksum. AliExpress · Amazon	Garage (CH1), green LED (CH2), red LED (CH3)	REQUIRED	10 to 18 €
IO-2	1	FTDI FT232RL USB-to-serial adapter, TTL levels (3.3/5 V), pin headers USB 0403:6001 → /dev/jarvis-door. CTS# input with a weak internal pull-up (about 200 kΩ) : add IO-9. Not an RS-232 DB9 cable (see JRV-DWG-002). Amazon · Mouser (chip)	Door sensor input	RECOMMENDED	5 to 12 €
IO-3	1	Wired magnetic reed contact for garage door, NO "Wide gap" model for sectional doors (floor or surface mounted). Amazon	Open/closed state: prevents "open" from closing the door	RECOMMENDED	10 to 25 €
IO-4	1	2-pair alarm cable 0.22 mm² (10 to 20 m) Reed → FT232RL. Twisted pair, away from 230 V. Amazon	Sensor link	RECOMMENDED	5 to 12 €
IO-5	1	2-conductor low-voltage cable, 2 × AWG 22 or larger Relay CH1 → terminal F (external pulse input) of the Novomatic 200; AWG 22 per the Novoferm manual (fig. 13a). Amazon	Garage pulse command	REQUIRED	3 to 8 €
IO-6	2	12 V LED indicators Ø 16/22 mm IP65 (1 green, 1 red) Panel indicators with built-in LED. Amazon	GO / NO_GO signal visible from outside	REQUIRED	5 to 12 €
IO-7	1	12 V DC 1 A power supply Enclosed, CE-certified adapter. Amazon	Powers the LEDs via CH2/CH3	REQUIRED	8 to 15 €
IO-8	1	IP55 junction box + lever terminal blocks (Wago 221 type) For the low-voltage connections. Amazon · Amazon	Protection of the connections	REQUIRED	8 to 20 €
IO-9	1	External pull-up resistor, a few kΩ, CTS# to VCCIO Recommended in docs/hardware/ftdi-ft232rL/ for a long sensor cable (with filtering).	Reliable door sensor level	RECOMMENDED	0 to 1 €

2.6 Power and miscellaneous

Ref.	Qty	Item / specification	Role	Status	Indicative price
PWR-1	1	UPS 600 to 850 VA Keeps the mini-PC and the PoE injector running through short outages. Amazon · LDLC	Availability	OPTIONAL	60 to 110 €

Ref.	Qty	Item / specification	Role	Status	Indicative price
PWR-2	1	Surge-protected power strip If there is no UPS. Amazon	Protection	OPTIONAL	10 to 20 €
MISC-1	1	VESA mount or wall bracket for the Tiny Lenovo 75/100 mm VESA kit or shelf. Amazon	Clean installation, ventilation	OPTIONAL	0 to 20 €
MISC-2	1	DisplayPort/VGA monitor + USB keyboard (borrowed) Only to install the system.	Installation	OPTIONAL	existing

3 CAMERA SELECTION (CAM-1)

Model	Sensor / zoom	Integration	Power	Remarks	Indicative price
Axis Q6078-E	4K 2160p (3840×2160)	ONVIF + VAPIX API	High PoE 60 W midspan supplied; 16 W typical, 51 W max. RJ45 10/100 only.	Most faithful ONVIF implementation, the most expensive.	≈ 2,500 to 4,000 €
Axis Q6075-E END OF LIFE	HDTV 1080p, 40× optical zoom	ONVIF Profiles G, S, T + VAPIX	IEEE 802.3bt, max. 51 W; High PoE 60 W midspan supplied. PoE+ 802.3at not enough.	End of life: last order date May 10, 2026, service/RMA until May 10, 2032. Official replacement per Axis: AXIS Q6086-E .	≈ 2,500 to 4,000 €
Hikvision DS- 2DF8C842IXS- AEL(T5)	8 MP (3840×2160), 42× zoom, IR 500 m	secondary stream /Streaming/Channels/1 02	24 V AC (62 W max.) or IEEE 802.3bt PoE (51 W max.)	Good value for money; IP67, about 9.6 kg.	≈ 1,200 to 1,900 €
Dahua SD6AL445XA- HNR	4 MP (2560×1440), 45× zoom, starlight + laser	secondary stream subtype=1	36 V DC / 2.23 A or Hi-PoE (36 W max.); no IEEE 802.3at/bt claim in the datasheet	Equivalent to Hikvision.	≈ 900 to 1,500 €
Reolink Argus PT (owner's camera) NOT SUITABLE	5 MP, fixed lens 90°, no optical zoom	no RTSP / ONVIF standalone	21.6 Wh battery + 3 W solar panel, Wi-Fi; no PoE	Not suitable without a Reolink Home Hub: through the Hub, RTSP sessions last 5 min at most and PTZ is not usable by Jarvis (ptz.enabled: false, fixed-camera mode).	already owned

NOTE

4K is not needed: the analysis runs on the 720p secondary stream and the optical zoom compensates. Check the exact resolution and the power class of the purchased model on the manufacturer's datasheet. The documents and key points of each camera are in docs/hardware/.

CAUTION

Reolink Argus PT: battery-powered Wi-Fi camera with no RTSP, RTMP or ONVIF when used standalone. It is usable only as a degraded, experimental source through a Reolink Home Hub (extra purchase), with `ptz.enabled: false`, and suffers regular blind gaps (5-minute sessions, wake-up of up to about 20 s). See docs/hardware/reolink-argus-pt/README.md.

4 COST SUMMARY AND POINTS OF ATTENTION

4.1 Cost summary

Item	Indicative amount
Required, camera excluded	172 to 591 €
PTZ camera (CAM-1)	900 to 4,000 €
Recommended (dedicated network, audio, door sensor)	108 to 205 €
Options (UPS, surge protection, USB extension...)	112 to 245 €
Total recommended configuration (required + recommended)	1,180 to 4,796 €

CAUTION

Ultralytics (AGPL-3.0), InsightFace and openWakeWord (non-commercial models) suit **private** use. Resale or a commercial service requires other models or commercial licenses.

4.2 Not counted

Existing equipment: **Novoferm Novomatic 200** door operator with its photocells and remote control, LAN router / switch, installation monitor and keyboard. Labor and small supplies (wall plugs, conduits, cable ties) are not included.

4.3 Hardware points of attention

- **CPU:** check with `lscpu` (4 cores and the `avx2` flag). The budget of 6 analyses per second assumes 4 cores; with an i5-4570T (2 cores), aim at 4 analyses per second and `imgsz`: 416.
- **No GPU card possible** on the Tiny (no PCIe); the mini-PCIe slot holds the Wi-Fi card. The architecture stays 100 % CPU, with VAAPI H.264 decoding.
- **USB ports:** 2 × USB 3.0 at the front (one always powered), 3 × USB 2.0 at the back. The BOM uses 4 of them (see the port allocation in JRV-DWG-002).
- **Door sensor:** the FT232R internal pull-up (about 200 kΩ) is weak; fit the external pull-up IO-9.
- **Novomatic terminal F:** its voltage is not documented by Novoferm; measure it before wiring.
- **Ventilation:** the Tiny gets hot under continuous load. Do not enclose it, dust it, watch sensors; install it indoors (garage), not outdoors.

5 SOFTWARE AND MODELS

No license cost for household use.

Component	Role	License	Remarks
Ubuntu Server 26.04 LTS	operating system	free (GPL and others)	reference platform; 24.04 accepted only with <code>jarvis_allow_ubuntu_2404</code>
Python 3.11 (via uv)	runtime	PSF	3.12 not usable (tflite-runtime, piper-phonemize wheels)
OpenVINO	YOLO inference on CPU	Apache-2.0	—
Ultralytics YOLO11n + ByteTrack	person detection and tracking	AGPL-3.0	personal use OK; distribution or commercial service = Ultralytics license
InsightFace buffalo_s (SCRFD + ArcFace) on ONNX Runtime	faces	code MIT, ONNX Runtime MIT; models: non-commercial research	non-commercial household use
openWakeWord "hey jarvis"	wake word	code Apache-2.0; models CC BY-NC-SA 4.0	non-commercial
Porcupine (option "jarvis")	wake word	proprietary, free key for personal use	Picovoice account
Vosk + vosk-model-small-fr-0.22	speech recognition (French commands)	Apache-2.0	—
Piper + voice fr_FR-siwis-medium	speech synthesis (French voice)	GPL-3.0 (piper-tts $\geq$ 1.3); voice: see its model card	personal use OK
SpeechBrain ECAPA-TDNN (option)	speaker verification	Apache-2.0	installs CPU torch ($\approx$ 700 MB)
FastAPI , Uvicorn, argon2-cffi, nginx , SQLite	web API, TLS, storage	MIT / BSD / public domain	—
Anubis	anti-bot proof-of-work proxy	MIT	pinned version and checksum (Ansible <code>anubis</code> role)

6 SOFTWARE BILL OF MATERIALS (SBOM)

Generated by `scripts/jarvis-gen-sbom.py` from the Python environment actually installed (`cyclonedx-py`), completed with the AI models (SHA-256 of each file) and the system components. Files: `docs/sbom/jarvis-sbom.cdx.json` (machine format, validated against the CycloneDX 1.6 schema, importable into Dependency-Track or readable by Grype/Trivy for vulnerability scanning) and `docs/sbom/jarvis-sbom.csv` (human-readable). **126 components.**

6.1 Components by type

CycloneDX type	Count
library	118
machine-learning-model	5
application	1
operating-system	1
platform	1

6.2 Components by license family

Family	Count
permissive (MIT, BSD, Apache, PSF...)	110
strong copyleft (GPL/AGPL)	6
weak copyleft (LGPL/MPL)	4
undeclared	4
non-commercial	2

6.3 Licenses to watch (everything that is not permissive)

Component	Version	Type	Declared license	Family
ffmpeg	4.4.2-0ubuntu0.22.04.1	application	LGPL-2.1-or-later	weak copyleft (LGPL/MPL)
certifi	2026.7.22	library	MPL-2.0 ; Mozilla Public License 2.0 (MPL 2.0)	weak copyleft (LGPL/MPL)
easydict	1.13	library	LGPL-3.0 ; LGPL-3.0-only	weak copyleft (LGPL/MPL)
hidapi	0.15.0	library	GPL-3.0-only ; BSD License	strong copyleft (GPL/AGPL)
multidict	6.9.1	library	UNDECLARED	undeclared
piper-tts	1.8.0	library	GPL-3.0-or-later	strong copyleft (GPL/AGPL)
protobuf	7.36.2	library	UNDECLARED	undeclared
torchvision	0.29.0+cpu	library	UNDECLARED	undeclared
tqdm	4.70.1	library	MPL-2.0 AND MIT	weak copyleft (LGPL/MPL)
ultralytics	8.4.165	library	AGPL-3.0 ; AGPL-3.0-or-later	strong copyleft (GPL/AGPL)
ultralytics-platform	0.1.70	library	AGPL-3.0-only	strong copyleft (GPL/AGPL)
ultralytics-thop	2.2.1	library	AGPL-3.0 ; AGPL-3.0-or-later	strong copyleft (GPL/AGPL)
insightface-buffalo_s	buffalo_s (insightface 0.7)	machine-learning-model	InsightFace pretrained models: non-commercial research only	non-commercial

Component	Version	Type	Declared license	Family
openwakeword-hey_jarvis	0.1	machine-learning-model	CC-BY-NC-SA-4.0	non-commercial
yolo11n	11 (export OpenVINO FP32)	machine-learning-model	AGPL-3.0-only	strong copyleft (GPL/AGPL)
ubuntu	22.04	operating-system	UNDECLARED	undeclared

NOTE

The delivered SBOM was produced on the test environment. Regenerate it on the installed machine (`sudo python3 scripts/jarvis-gen-sbom.py`) to reflect the exact OS, FFmpeg and nginx in place, and after every dependency update.

APPENDIX A GLOSSARY

Technical terms and abbreviations used in this document.

802.3af / at / bt

IEEE Power over Ethernet standards: 15.4 W (af, "PoE"), 30 W (at, "PoE+"), 60 to 90 W at the source (bt, "PoE++", classes 5 to 8). The PTZ cameras of the project draw up to 51 W and need 802.3bt or the vendor's Hi-PoE injector.

AEC

Acoustic Echo Cancellation: removal of the loudspeaker signal from the microphone signal. Performed here by the XMOS XVF-3000 chip of the ReSpeaker.

AGPL-3.0

GNU Affero General Public License v3: strong copyleft license that also covers use over a network. License of Ultralytics YOLO.

Anubis

Proof-of-work reverse proxy that keeps crawlers, AI agents and scrapers away from the web UI (instance anubis@jarvis on 127.0.0.1:8923).

API (REST)

HTTP programming interface in which each resource (person, sighting...) has a URL and is handled with the GET, POST, PATCH and DELETE verbs.

ArcFace

Neural network and loss function (additive angular margin) that turn an aligned face into a 512-dimension vector, compared by cosine similarity.

AVX2

256-bit vector instruction set of Intel processors since Haswell, used by OpenVINO and ONNX Runtime. Missing from the Celeron and Pentium parts of that generation.

BOM

Bill of Materials: exhaustive list of the hardware components, with quantities and references.

ByteTrack

Multi-object tracking algorithm that associates detections from one frame to the next by IoU, in two passes (high then low confidence), with Kalman filter prediction.

CH340

WCH USB-to-serial converter chip fitted to the LCUS relay boards.

CTS / DTR

Control lines of a serial link: CTS (Clear To Send) is an input, DTR (Data Terminal Ready) an output. CTS is used as the logic input of the door sensor.

CycloneDX

OWASP SBOM standard (JSON/XML format). Version used: 1.6.

Dry contact (potential-free)

Electrical contact that supplies no voltage: it only closes or opens a circuit powered by the connected equipment. The only type allowed on terminal F of the Novomatic 200.

ECAPA-TDNN

Speaker verification neural network (SpeechBrain) that produces a 192-dimension voiceprint.

F/UTP

Twisted-pair cable with an overall foil screen (Foiled / Unshielded Twisted Pairs).

FFmpeg

Audio/video decoding and encoding library and tool, used for the RTSP stream and for converting the voice recordings.

FT232R / FT232RL

FTDI USB-to-serial converter chip with TTL levels; its CTS# pin reads the door sensor. Its internal pull-up (about 200 kΩ) is weak.

H.264 / H.265

Video compression standards (AVC / HEVC). The analyzed stream must be H.264, the only one the Intel HD 4600 decodes in hardware.

Hi-PoE / High PoE

Proprietary high-power PoE (Hikvision, Dahua; "High PoE" at Axis), close to 802.3bt.

Injector (PoE midspan)

Unit that adds power to an Ethernet cable between a network device and a powered device.

InsightFace

Open-source face analysis library (SCRFD detection, ArcFace recognition).

IP55 / IP65 / IP66 / IP67

Ingress protection ratings (IEC 60529): 6 = dust-tight; 5 = water jets, 6 = powerful jets, 7 = temporary immersion.

IR

Infrared: invisible lighting that gives a black-and-white image at night.

LCUS-4

4-relay board driven over USB through a CH340, serial frame A0 channel state checksum.

Lever terminal block

Screwless connector with a lever (Wago 221 type) for 0.14 to 4 mm² conductors.

NO / NC / COM

Relay terminals: COM common, NO normally open (closed when the relay is energized), NC normally closed.

ONNX / ONNX Runtime

Open AI model format and optimized inference engine (Microsoft).

ONVIF

IP camera interoperability standard (discovery, streaming, PTZ). Profile S: video and PTZ.

OpenVINO

Intel inference engine optimized for its processors.

openWakeWord

Open-source wake-word detector (ONNX models).

Piper

Offline neural text-to-speech engine (VITS exported to ONNX).

PoE

Power over Ethernet: electrical power carried by the network cable.

Porcupine

Commercial wake-word detector from Picovoice (free key for personal use).

PTZ

Pan-Tilt-Zoom: camera motorized in pan, tilt and zoom.

Pull-up

Resistor that pulls a logic input to the high state when nothing drives it low.

Pulse

Short closure (500 ms) of the relay contact, equivalent to pressing the push button.

Reed switch

Switch with flexible blades operated by a magnet; used as a door opening sensor.

RTSP

Real Time Streaming Protocol (RFC 2326/7826): streaming protocol of the camera video.

SBOM

Software Bill of Materials: exhaustive inventory of the software components, versions and licenses.

SCRFD

Fast InsightFace face detector (Sample and Computation Redistribution).

SQLite (WAL)

Database embedded in a single file; Write-Ahead Log mode allows concurrent reads and a write.

TLS / HTTPS

Encryption of web traffic (Transport Layer Security).

Track

Sequence of detections of the same person over time, identified by a number (ByteTrack).

TTS

Text-To-Speech, speech synthesis.

uv

Python package and environment manager (Astral).

VAAPI

Video Acceleration API: hardware video decoding on Intel GPUs.

Vosk

Offline speech recognition engine built on Kaldi.

Wake word

Word or phrase ("hey jarvis") that activates command listening.

YOLO

You Only Look Once: family of single-pass object detectors; YOLO11n is the nano variant.

APPENDIX B BIBLIOGRAPHY

Further reading. Titles are given as published; ISBNs are those of the edition cited and are omitted when they could not be verified.

B.1 Books in English

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